How to simulate data

review of some simulation settings in causal inference

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My settings

N obs data, M rct data; d covariates, within which c is confounder
[do we need to center the response?]
1. $e(x)$, overlap, well-specified or not
2. N/M
3. c/d, how many irrelevant covariates
4. d(c)/N, high-dimensional settings or not
5. sample size, how much information

Resources: `https://www2.stat.duke.edu/~fl35/OW/OW_AOAS_supp.pdf`

pre-treatment covariate X

types of covariates

If we have two models and homogeneous effect:

$$m_0(X) = E(Y(0)|X)$$

$$\tau(X) = \tau = E(Y(1) - Y(0))$$
$$e(X) = E(Z = 1|X)$$

- X predictive of outcome but not trt (prognsotic covariates)
- X predictive of trt but not outcome (instrumental variables)
- X predictive of both trt and outcome (confounders)

What about when we have three models?

$$m_0(X) = E(Y(0)|X) \text{ (or main effect } m(X) = 2^{-1}E[Y(0) + Y(1) \mid X]\text{)}$$

$$\tau(X) = E(Y(1) - Y(0)|X)$$
$$e(X) = E(Z = 1|X)$$

distribution of X

any restrictions?(like bounded...)

Simulation settings

(Li, Thomas, and Li 2019) (Yang, Thomas, and Li 2026) (how much overlap, i.e. how strong tails of $e(x)$)

- X

$$V = (V_1, \ldots, V_6)^\top \sim \mathcal{N}(0, \Sigma_{6\times 6}), \quad \Sigma_{jj} = 1, \quad \Sigma_{jk} = 0.5 \ \ (j \neq k),$$

$$X_j = \begin{cases} V_j, & j = 1, 2, 3 (\text{continuous}), \\ \mathbf{1}_{\{V_j < 0\}}, & j = 4, 5, 6 (\text{binary}). \end{cases}$$

- $Z \mid X$ (vary γ from 1 to 4 to reflect increasingly sparse overlap between groups)

$$e(X) = \{1 + \exp[-(\alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 + \alpha_4 X_4 + \alpha_5 X_5 + \alpha_6 X_6)]\}^{-1}$$

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6) = \gamma \times (0.15, 0.3, 0.3, -0.2, -0.25, -0.25)$$

- $Y \mid X$ (a homogenous and additive treatment effect Δ)

$$E(Y|Z, X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \Delta Z$$

$$\beta = (0, -0.5, -0.8, -1.2, 0.8, 0.8, 1)$$

(so that subjects with lower outcomes are more likely to receive the treatment)

$$Y \sim N(E(Y \mid X), 1.5)$$

- model misspecification

i) wrong functional form (mistaking X_3 by X_3^2) and ii) omitting variables (X_3 and X_4).

(Rubin 1973) [ATT, homogeneous/parallel, matching]

- $X \mid Z = z$

$$X \mid Z = 1 \sim \mathcal{N}(\eta_1, \sigma_1^2), \quad X \mid Z = 0 \sim \mathcal{N}(\eta_2, \sigma_2^2)$$

  where

$$\eta_1 = -\eta_2, \ (\sigma_1^2 + \sigma_1^2)/2 = 1$$

  - variance ratio: $\sigma_1^2/\sigma_2^2 \in \{1/2, 1, 2\}$ (a ratio of 1/2 is more "favorable to match" than 1 because the target group is more concentrated than the reservoir group).
  - mean discrepancy: $B = 2\eta_1 \in \{1/4, 1/2, 3/4, 1\}$, which is the standard deviation between the means of X

- $Y(z) \mid X$, response surface $V(x)$ (Robustness to poor overlap)

$$Y_z(x) = \alpha_z + V(x), \ V(0) = 0, \ \alpha_1 - \alpha_0 = \tau$$

$$V(x) = exp(ax), \ a \in \{1, 1/2, -1/2, -1\}$$

  Since $\eta_1 > \eta_2$, compared with $exp(x/2)$, the response surfaces $exp(x)$ is "less robust to poor overlap" because it is increasing most rapidly for large X which is where close matches are difficult to find, thus this is "less favorable to matching" than $exp(x/2)$.

- sample size of treatment N and control groups rN

$$r \in \{2, 3, 4\}$$

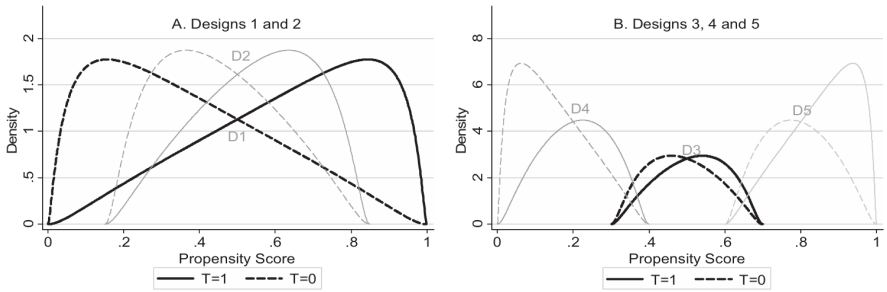
(Frölich 2004) (Busso, DiNardo, and McCrary 2014) (ATE, compare overlap's influence on weighting and matching estimators)

- X one-dimensional from Johnson S_B distribution, support [0,1]
- $Z \mid X$ propensity score

$$e(X) = \alpha + \beta\Lambda(\sqrt{2}X) = k(\Lambda(\sqrt{2}X))$$

$$k(X_i) \equiv \beta_0 + \sum_{\ell=1}^{L} \left\{\beta_\ell^S \sin(\ell x_{i,n}) + \beta_\ell^C \cos(\ell x_{i,n})\right\}$$

FIGURE 1.—FROLICH PROPENSITY SCORE CONDITIONAL DENSITIES

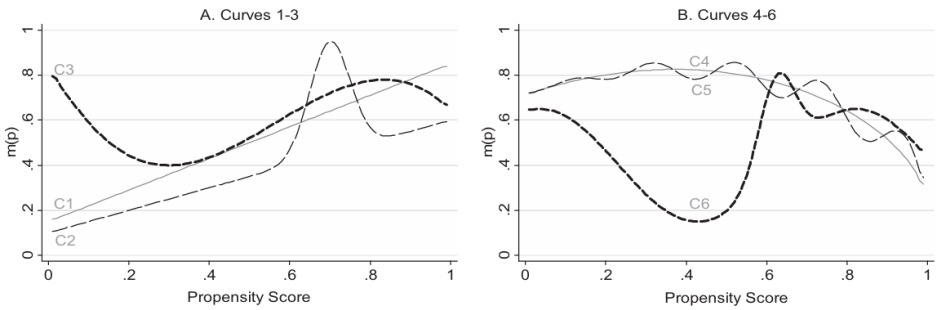


Note: Each selection equation design is made following Frölich (2004) designs. Equation (5) in section III is specified as follows: Design 1 ($\alpha = 0, \beta = 1$), design 2 ($\alpha = 0.15, \beta = 0.7$), design 3 ($\alpha = 0.3, \beta = 0.4$), design 4 ($\alpha = 0, \beta = 0.4$), design 5 ($\alpha = 0.6, \beta = 0.4$).

- $Y \mid X, e(X)$

$$Y_i(1) = Y_i(0) = m(\Lambda(\sqrt{2}X)) + \sigma\varepsilon_i$$

FIGURE 2.—FROLICH OUTCOME CURVES



Note: Each panel displays the conditional expectation of $Y(0)$ given the propensity score p for the curve in question. See the text for details and Frölich (2004, table A1).

(Wager and Athey 2018) (how $\tau(x)$ related with X, HTE)

Main effect: $m(x) = 2^{-1}\mathbb{E}[Y^{(0)} + Y^{(1)} \mid X = x]$
Treatment effect: $\tau(x) = \mathbb{E}[Y^{(1)} - Y^{(0)} \mid X = x]$

$$E(Y|X, Z) = m(X) + (Z - 0.5)\tau(X)$$

- X (CRF need bounded covariate)

$$X \sim U([0,1]^d)$$

- $Y \mid X$, $Z \mid X$

$$Y^{(0/1)} \sim N(E[Y^{(0/1)} \mid X], 1)$$

  - non-random assignment $e(X) = \frac{1}{4}(1 + \beta_{2,4}(X_1))$, non-zero main effect, homogeneous

$$m(X) = 2X_1 - 1, \quad \tau(X) = 0$$

  - random assignment $e(X) = 0.5$, zero mean effect $m(X) = 0$, heterogeneous

$$\tau(x) = \varsigma(X_1)\varsigma(X_2), \quad \varsigma(x) = 1 + \frac{1}{1 + e^{-20(x-1/3)}}$$

  - random assignment $e(X) = 0.5$, zero mean effect $m(X) = 0$, heterogeneous[1]

$$\tau(x) = \varsigma(X_1)\varsigma(X_2), \quad \varsigma(x) = \frac{2}{1 + e^{-12(x-1/2)}}$$

- evaluation: We evaluate performance in terms of expected mean-squared error for estimating $\tau(X)$ at a random test example X, as well as expected coverage of $\tau(X)$ with a target coverage rate of 0.95.

[1] this setting for: weakness of nearest neighbor approaches(random forests) they can fill the valleys and flatten the peaks of the true $\tau(x)$ function, especially near the edge of feature space

(Nie and Wager 2021) [heterogeneity, many baseline estimators]

metric: the oracle test set mean-squared error

baseline methods

- S-learner
- T-learner
- X-learner
- U-learner
- RS-learner

simulation settings

$$Y_i = b(X_i) + (W_i - 0.5)\tau(X_i) + \sigma\epsilon_i$$

- A: Difficult Nuisance, Easy Treatment Effect, Heterogeneity
  - $X \sim \mathrm{U}(0,1)^d$
  - $e(X) = \mathrm{trim}_{0.1}\{\sin(\pi X_{i1}X_{i2})\}$ (Non-linear propensity score)
  - $b(X) = \sin(\pi X_{i1}X_{i2}) + 2(X_{i3} - 0.5)^2 + X_{i4} + 0.5X_{i5}$ (Friedman function)
  - $\tau(X) = (X_{i1} + X_{i2})/2$
- B: Randomized Trial, Heterogeneity
  - $X \sim N(0, I_{d\times d})$
  - $e(x) = 1/2$
  - $b(X) = \max\{X_{i1} + X_{i2}, X_{i3}, 0\} + \max\{X_{i4} + X_{i5}, 0\}$
  - $\tau(X) = X_{i1} + \log\{1 + \exp(X_{i2})\}$
- C: Easy Propensity, Difficult Baseline, No heterogeneity
  - $X \sim N(0, I_{d\times d})$
  - $e(X) = \{1 + \exp(X_{i2} + X_{i3})\}^{-1}$
  - $b(X) = 2\log\{1 + \exp(X_{i1} + X_{i2} + X_{i3})\}$ (Strong confounding but complex baseline?)
  - $\tau(X) = 1$
- D: Unrelated Treatment and Control Arms(no benefit to joint learning), Heterogeneity
  - $X \sim N(0, I_{d\times d})$
  - $e(X) = \{1 + \exp(-X_{i1}) + \exp(-X_{i2})\}^{-1}$
  - $b(X) = [\max\{X_{i1} + X_{i2} + X_{i3}, 0\} + \max\{X_{i4} + X_{i5}, 0\}]/2$
  - $\tau(X) = \max\{X_{i1} + X_{i2} + X_{i3}, 0\} - \max\{X_{i4} + X_{i5}, 0\}$

(Wager 2024) [HTE, compare lasso-based R-learner and T-learner]

- X

$$X \sim \mathcal{N}\left(0, I_{10\times 10}\right)$$

- $Z \mid X$, $Y \mid X$(regularization-induced confounding)[2]

$$e(X) = 1/\left(1 + e^{-(X_2+X_3)}\right)$$

$$Y(z) = 2\log\left(1 + e^{X_1+X_2+X_3}\right) + zI\left(X_2 + X_3 \geq 0\right) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1)$$

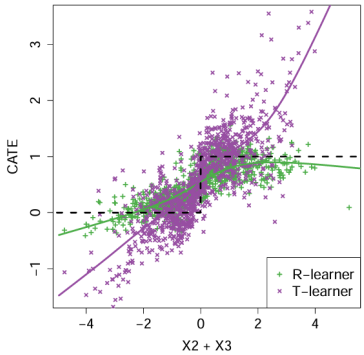


Figure 1: regularization-induced-confounding

[2]this setting for: units for which $X_2 + X_3$ is large are simultaneously more likely to be treated, have a larger baseline effect whether or not they get treated, and have a larger treatment effect.

(Diamond and Sekhon 2013) (Lee, Lessler, and Stuart 2010)

Setting 1

- X ten covariates $(X_k, k = 1, ..., 10)$: four confounders associated with both treatment and outcome, three treatment predictors(instruments[3]), and three outcome predictors. Six covariates $(X_1, X_3, X_5, X_6, X_8, X_9)$ are binary, whereas four (X_2, X_4, X_7, X_{10}) are standard normal.

- $Z \mid X$ (change the degree of linearity and additivity in the true propensity score model) (the average probability of treatment assignment at the average value of the covariates is ≈ 0.5)

  A: Additivity and linearity (mean effects only)

  B: Mild nonlinearity (one quadratic term)

  C: Moderate nonlinearity (three quadratic terms)

  D: Mild nonadditivity (three two-way interaction terms)

  E: Mild nonadditivity and nonlinearity (three two-way interaction terms and one quadratic term)

  F: Moderate nonadditivity (ten two-way interaction terms)

  G: Moderate nonadditivity and nonlinearity (ten two-way interaction terms and three quadratic terms)

- $Y \mid X$ (homogeneous effect, linear model)

$$Y = \gamma Z - 3.85 + 0.3X_1 - 0.36X_2 - 0.73X_3 - 0.2X_4 + 0.71X_8 - 0.19X_9 + 0.26X_{10}$$

Setting 2

- X based on the LaLonde(1986) data. Some of the baseline variables are discrete, and others contain pointmasses and skewed distributions. None of the covariates,as they are based on real data, have ellipsoidal distributions[?].

- $Z \mid X$ (not correctly specified propensity model)

$$\begin{aligned}\pi_i = \text{logit}^{-1}[&1 + .5\hat{\mu} + .01\,\text{age}^2 - .3\,\text{educ}^2 \\ &- .01\log(\text{re74} + .01)^2 + .01\log(\text{re75} + .01)^2],\end{aligned} \tag{4}$$

  This propensity score is a mix of the estimated propensity score plus extra variables, so that the propensity model estimated in the Monte Carlo samples is badly misspecified[4].

- $Y \mid X$

$$\begin{aligned}Y =& 1000\,T + .1\exp[.7\log(\text{re74} + .01) \\ &+ .7\log(\text{re75} + 0.01)] + \epsilon,\end{aligned}$$

[3] In practice, instruments should not be included in the propensity score when assuming selection on observed variables (and not in an OLS regression model if that is used)[?]

[4] GenMatch is able to obtain better levels of balance without requiring the analyst to correctly specify the propensity score.

(Wang and Shah 2024) [ATE, high-dimensional confounding]

- X
  - Toeplitz $X_i \sim N(0, \Sigma)$, $\Sigma \in R^{p \times p}$, $\Sigma_{i,j} = 0.9^{|i-j|}$
  - Exponential decay $X_i \sim N(0, \Sigma)$, $\Sigma \in R^{p \times p}$, $(\Sigma^{-1})_{i,j} = 0.9^{|i-j|}$, and normalize diagonal entries to 1
- $Y \mid X$ (No restriction on the sparsity of outcome model, any complex model)

$$Y_i = b(X_i) + Z_i \Delta(X_i) + \varepsilon_i$$

6.1.1.2. *Response function* $b(\cdot)$. We consider the following two forms of the response function:

- *Dense linear function*: We set $b(x) = x^\top \beta$, where we generate $\beta \in \mathbb{R}^p$ by first randomly selecting 50 components and then independently assigning each selected entry a random value uniformly sampled from $[0, 1]$. All other entries are set to 0 and finally β is normalised such that $\|\beta\|_2 = 2$.
- *Nonlinear function*: We generate $\beta \in \mathbb{R}^p$ as above and then set $b(x) = 3\tilde{x}^\top \beta$, where $\tilde{x}_j = 2\{1 + \exp(-x)\}^{-1} - 1$.

6.1.1.3. *Heterogeneous treatment effect function* $\Delta(\cdot)$. We consider the following two forms of heterogeneous treatment effects:

- *Dense linear function*: We generate a vector $\delta \in \mathbb{R}^p$ in the same fashion as β above, but normalise $\|\delta\|_2 = 1$; we then set $\Delta(x) = x^\top \delta$.
- *Nonlinear function*: We first generate $\delta \in \mathbb{R}^p$ as above and then set $\Delta(x) = \{1 + \exp(x^\top \delta)\}^{-1} - 0.5$.

- $Z \mid X$, $P(Z = 1 \mid X) = logis(X_i^T \gamma)$ (need sparsity of propensity model)

  Given the randomly generated β, randomly select $s = 5, 20, 50$ indices corresponding to nonzero entries in β, set a random value uniformly sampled from [0,1] and the remaining components set to zero. normalize such that $||\gamma||_2 = 1$.

(Athey, Imbens, and Wager 2018) [high-dimensional, cluster?]

(Zhao 2019) [CBPS]

Setting 1: Low-dimensional covariates (choose a function class to balance covariate on)

- X, $X_i \sim N(0, I_5)$
- $Z \mid X$ (random logistic function with kernel)
  - Gaussian kernel
  - Laplace kernel
  - polynomial kernel
- $Y \mid X$ (random function also generated by Gaussian process

$$Y_i(0) = Y_i(1) = g_0(X_i) + \varepsilon_i$$

Setting 2: High-dimensional covariates

- X, $X_i \in R^{100} \sim N(0, \Sigma)$, $\Sigma_{ij} = 0.5^{|i-j|}$
- $Z \mid X$,

$$P(Z = 1 \mid X) = \rho X^T \theta, \ \rho = 1, 2, \ \theta = (1/\sqrt{s_t}, ..., 1/\sqrt{s_t}, 0, ..., 0), \ s_t = 5, 100$$

- $Y \mid X$

$$Y_i(0) = Y_i(1) = X_i^T \beta + \varepsilon_i, \ \beta = (1/\sqrt{s_y}, ..., 1/\sqrt{s_y}, 0, ..., 0), \ s_y = 5, 20, 100$$

Reported values:

> Averaging over 1000 simulations, we report in Table 3 the root-mean-square error of the estimators (RMSE), the absolute bias (Bias), the estimated maximum bias as described in Section 5.5 which uses the EigenPrism method of Janson, Foygel Barber and Candès [30] to estimate $\|\boldsymbol{\beta}\|_2$ (Max Bias), coverage of the 95% confidence interval ignoring covariate imbalance as in Athey et al. [2] (CI), coverage of the honest 95%-confidence interval (27) (Honest CI) and the ratio of the length between the two confidence intervals (CI Ratio).

(D'Amour et al. 2021) (strict overlap means gaussian covariates should perfectly balance)

- If X is one-dimensional and follows a Gaussian distribution under both P_0 and P_1, **strict overlap implies that $P_0 = P_1$, or that the covariate is perfectly balanced**. Similar results can be derived when $X_{1:p}$ is multi-dimensional Gaussian. Thus, for Gaussianly distributed covariates, the implications of strict overlap are so strong that they are uninteresting.

- simulation intuition
  - $P_1 = P_0 \sim N(\mu, \Sigma)$
  - $P_1 \neq P_0,\ P_1 \sim N(\mu_1, \Sigma_1),\ P_0 \sim N(\mu_0, \Sigma_0)$
  - Truncated Normal
  - Uniform/Beta/..

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